Integration — Lesson 12 Test: Motion II — Acceleration & Gravity

10 questions · show your work in the box · use $g = 10\ m/s^2$ · calculators allowed · every answer needs UNITS

Name: ______________________________ Date: ______________ Score: ______ / 10

1. A sled starts from rest and accelerates at a steady $5\ m/s^2$. Write the velocity formula $v(t)$, and find the velocity after 6 seconds. 1 pt

2. A skateboarder is already moving at $4\ m/s$ and accelerates at $3\ m/s^2$. Write $v(t)$ and find the velocity after 2 seconds. 1 pt

3. A ball is dropped from a hot-air balloon. After 4 seconds, how fast is it moving and how far has it fallen? ($g = 10\ m/s^2$) 1 pt

4. A coin is dropped from a 45-meter rooftop. How long does it take to hit the ground, and how fast is it moving when it lands? 1 pt

5. An object's velocity is $v(t) = 2 + 6t$ m/s. Use an integral to find how far it travels in the first 3 seconds. 1 pt

6. A ball is thrown straight up at 40 m/s (up = positive, $a = -10$ m/s²). Find the time it reaches its peak, the peak height, and the time it lands back at its starting height. 1 pt

7. A car moving at 30 m/s brakes with constant acceleration -6 m/s². How long does it take to stop, and how far does it travel while stopping? 1 pt

8. An acceleration graph is a staircase: $a = 2 \text{ m/s}^2$ from $t = 0$ to $t = 3$ s, then $a = 4 \text{ m/s}^2$ from $t = 3$ to $t = 5$ s. The object's starting velocity is 5 m/s. Use rectangle areas to find the velocity at $t = 5$. (Careful with the width of the second rectangle!) 1 pt

9. Derive the formula $v = v_0 + at$ by integrating a constant acceleration a. Show the integral and explain why the starting velocity v_0 must be added. 1 pt

10. A model rocket on a track starts from rest. Phase 1: $a = 8 \text{ m/s}^2$ for 3 seconds. Phase 2: the engine cuts out and it coasts at constant velocity for 5 more seconds. Find its velocity after phase 1 and the total distance covered over all 8 seconds. 1 pt